

National University of Computer and Emerging Sciences



**Confidence-Aware Self-Prompting for Robust Polyp
Segmentation Under Image and Domain Shifts**

F26-13

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Problem Statement

Colonoscopy is widely used for the examination of the gastrointestinal tract and for the detection and assessment of colorectal polyps. Polyps can vary considerably in size, shape, color, and appearance, making their detection and precise delineation challenging in colonoscopy images. Accurate localization and segmentation of polyps can assist computer-aided analysis and reduce the possibility of missed lesions.

Deep learning methods have been increasingly used for **automatic polyp detection and segmentation**. Object detection models can identify the approximate location of a polyp using a bounding box, while segmentation models can provide a more precise pixel-level representation of the detected region. The selected YOLO-SAM2 approach combines these two stages by using **YOLOv8** for polyp detection and using the resulting bounding box as a prompt for SAM 2 to generate the final segmentation mask.

The performance of this pipeline depends partly on the quality of the **bounding box** generated by the detection model. An inaccurate or uncertain detection can provide a

weak prompt to the segmentation model, which may result in an inaccurate segmentation mask. This creates a need to investigate whether the automatically generated prompt can be improved before it is passed to the segmentation model.

Therefore, this project investigates whether a **confidence-aware prompt refinement strategy can improve the performance of the YOLO-SAM2 pipeline**. The proposed method will use the confidence of YOLO detections to identify uncertain cases and refine the corresponding bounding-box prompt before passing it to SAM 2. The proposed approach will be compared with the original YOLO-SAM2 pipeline to determine whether prompt refinement improves polyp segmentation accuracy and robustness.

Datasets

The project will use two publicly available colonoscopy image datasets. Kvasir-SEG will be used as the primary dataset for implementation and evaluation, while CVC-ClinicDB will be used as an external dataset to evaluate the performance of the proposed method on a different collection of colonoscopy images.

Dataset	Role in Study	Key Features	Source
Kvasir-SEG	Primary training and evaluation dataset	Contains 1,000 colonoscopy images with corresponding pixel-level polyp segmentation masks and bounding-box information. The masks were manually annotated and verified by an experienced gastroenterologist.	Simula Research Laboratory: https://datasets.simula.no/kvasir-seg/
CVC-ClinicDB	External evaluation dataset	Public colonoscopy image dataset containing polyp images and corresponding segmentation annotations. It will be used to evaluate the generalization of the baseline and proposed approaches on a different dataset.	Grand Challenge: https://polyp.grand-challenge.org/CVCClinicDB/

Relevant Papers

[1] Gaussian YOLOv3: An Accurate and Fast Object Detector Using Localization Uncertainty for Autonomous Driving Choi et al. (2019)

Shows that a detector's predicted localization uncertainty, derived from its confidence output, is a reliable indicator of bounding-box quality. Directly motivates using YOLO's confidence score to flag uncertain detections before they are used as a downstream prompt.

<https://arxiv.org/abs/1904.04620>

[2] Bounding Box Regression with Uncertainty for Accurate Object Detection He et al. (2019)

Introduces a method (KL Loss) for explicitly modelling uncertainty in bounding-box coordinates and using it to refine box predictions. Provides a concrete precedent for the type of confidence-driven box refinement proposed in this project.

<https://arxiv.org/abs/1809.08545>

[3] On the Robustness of Segment Anything Huang et al. (2023)

Empirically shows that SAM's segmentation quality is sensitive to the type of image corruption, notably blur, and to the prompt supplied to it. Supports the hypothesis that refining a low-quality prompt, rather than only improving the segmentation model, can improve robustness of the pipeline.

<https://arxiv.org/abs/2305.16220>

[4] Self-Prompting Polyp Segmentation in Colonoscopy using Hybrid Yolo-SAM 2 Model Mansoori et al. (2024)

The paper proposes a hybrid polyp segmentation pipeline in which YOLOv8 is used to detect polyps and the predicted bounding boxes are automatically used as prompts for SAM 2. This removes the need for manually providing prompts to the segmentation model. The paper evaluates the proposed approach on multiple colonoscopy image and video datasets. The YOLO-SAM2 implementation will serve as the baseline for this project.

<https://arxiv.org/abs/2409.09484>

Code Repository: https://github.com/sajjad-sh33/YOLO_SAM2

[5] SAM 2: Segment Anything in Images and Videos Ravi et al. (2024)

Introduces SAM 2, a promptable segmentation model designed for image and video segmentation. In this project, SAM 2 will be used to generate pixel-level polyp masks from the bounding-box prompts generated by the YOLO detection stage.

<https://arxiv.org/abs/2408.00714>

[6] Kvasir-SEG: A Segmented Polyp Dataset Jha et al. (2020)

Introduces the Kvasir-SEG dataset containing gastrointestinal polyp images with corresponding pixel-level segmentation masks and bounding-box information. The dataset provides the primary data source for training and evaluating polyp segmentation methods.

<https://arxiv.org/abs/1911.07069>

Research Hypothesis

The project will investigate the following hypothesis:

Using YOLO detection confidence to identify uncertain detections and refine their bounding-box prompts before passing them to SAM 2 will improve polyp segmentation accuracy and robustness compared with directly using the original YOLO-generated prompts.

The hypothesis will be tested by comparing the original YOLO-SAM2 pipeline with the proposed confidence-aware prompt refinement approach. The comparison will consider both segmentation performance and the behavior of the models on uncertain or difficult cases.

Project Scope

This project focuses on improving automatic polyp segmentation in colonoscopy images using a YOLO-SAM2 pipeline. The scope includes reproduction of the baseline implementation, development of a confidence-aware prompt refinement method, comparative evaluation, and testing on an external dataset.

5.1 Dataset Preparation

- Obtain and prepare the Kvasir-SEG dataset for the primary experiments.
- Use the available segmentation masks as ground-truth annotations for evaluating the predicted masks.
- Obtain and prepare CVC-ClinicDB for external evaluation.
- Apply the preprocessing and input preparation required by the selected YOLO-SAM2 implementation.

5.2 Baseline Implementation

- Set up and execute the existing YOLO-SAM2 implementation.
- Use YOLOv8 to detect the location of the polyp in each colonoscopy image.
- Use the YOLO-generated bounding box as the prompt provided to SAM 2.
- Generate the final pixel-level segmentation mask using SAM 2.
- Evaluate the baseline results against the available ground-truth masks.

5.3 Confidence-Aware Prompt Refinement

The proposed development will focus on the interaction between the detection and segmentation stages.

- Analyze the confidence score produced by YOLO for each detection.
- Identify cases where the detector is uncertain or where the predicted bounding box may not adequately cover the polyp.
- Develop a prompt refinement strategy for uncertain detections.
- Modify or refine the bounding-box prompt before passing it to SAM 2.

- Evaluate whether the refined prompt produces a more accurate segmentation mask.

5.4 Comparative Evaluation

The proposed approach will be compared with the original YOLO-SAM2 pipeline.

The main comparison will be:

Baseline:

YOLOv8 → Bounding Box → SAM 2 → Segmentation Mask

Proposed:

YOLOv8 → Confidence Analysis → Prompt Refinement → SAM 2 → Segmentation Mask

The segmentation results will be evaluated using appropriate metrics, including Dice coefficient and Intersection over Union (IoU). These metrics are commonly used for polyp segmentation and are also suggested for evaluation of the Kvasir-SEG dataset.

5.5 Cross-Dataset Evaluation

The baseline and proposed approaches will be evaluated on CVC-ClinicDB in addition to the primary Kvasir-SEG experiments.

This experiment will be used to examine whether the proposed confidence-aware prompt refinement approach maintains its effectiveness when applied to images from a different dataset.

5.6 Error and Ablation Analysis

The project will analyze cases where the proposed method improves, maintains, or decreases segmentation performance.

The analysis will focus on the effect of prompt refinement on uncertain detections and will include appropriate ablation experiments where possible. The results will be used to determine whether the observed changes are attributable to the proposed confidence-aware component.

Project Overview (Scope of Pipeline)

The project will cover the following high-level steps:

1. Dataset Preparation

- Download and prepare Kvasir-SEG as the primary dataset.
- Prepare the corresponding ground-truth segmentation masks and bounding-box information.
- Prepare CVC-ClinicDB for external evaluation.

2. Baseline Model Execution

- Set up the existing YOLO-SAM2 implementation.
- Execute YOLOv8 for automatic polyp detection.
- Use the predicted bounding box as the prompt for SAM 2.
- Generate and evaluate the resulting segmentation masks.

3. Confidence-Aware Prompt Refinement

- Analyze YOLO detection confidence.
- Identify uncertain detections.
- Apply the proposed bounding-box/prompt refinement strategy.
- Pass the refined prompts to SAM 2.

4. Comparative Evaluation

- Compare the original YOLO-SAM2 pipeline with the proposed approach.
- Calculate Dice and IoU for segmentation performance.
- Analyze differences between the baseline and proposed method.

5. External Dataset Evaluation

- Evaluate both approaches on CVC-ClinicDB.
- Compare performance between the primary and external datasets.
- Analyze whether the proposed method improves robustness across datasets.

6. Analysis and Reporting

- Perform error analysis on successful and unsuccessful cases.
- Conduct appropriate ablation/comparative experiments.
- Document the methodology, experimental results, limitations, and findings in the final technical report.

Expected Outcome

The project will produce a reproducible implementation of the YOLO-SAM2 baseline together with a confidence-aware prompt refinement component.

The main outcome will be a comparative analysis of the original and proposed pipelines. The experiments will determine whether refining uncertain YOLO-generated prompts improves the accuracy and robustness of polyp segmentation on the selected datasets.

The final project will include the implementation, experimental results, analysis of errors and limitations, and a technical report documenting the findings.

References

- [1] **Gaussian YOLOv3: An Accurate and Fast Object Detector Using Localization Uncertainty for Autonomous Driving — Choi et al. (2019)**

<https://arxiv.org/abs/1904.04620>

[2] Bounding Box Regression with Uncertainty for Accurate Object Detection — He et al. (2019)

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[3] On the Robustness of Segment Anything — Huang et al. (2023)

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[4] Self-Prompting Polyp Segmentation in Colonoscopy using Hybrid Yolo-SAM 2 Model — Mansoori et al. (2024)

<https://arxiv.org/abs/2409.09484>

[5] SAM 2: Segment Anything in Images and Videos — Ravi et al. (2024)

<https://arxiv.org/abs/2408.00714>

[6] Kvasir-SEG: A Segmented Polyp Dataset — Jha et al. (2020)

<https://arxiv.org/abs/1911.07069>

[7] Kvasir-SEG Dataset — Simula Research Laboratory <https://datasets.simula.no/kvasir-seg/>

[8] CVC-ClinicDB Dataset — Grand Challenge <https://polyp.grand-challenge.org/CVCClinicDB/>

[9] YOLO-SAM2 Official Code Repository https://github.com/sajjad-sh33/YOLO_SAM2